

GIRRAWEEEN 2006 EX 1 TRIAL

Total marks – 84

Attempt Questions 1 – 7

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (12 marks) Use a separate piece of paper Marks

- a) Find $\int_0^5 \frac{dx}{\sqrt{25-x^2}}$ 2
- b) Find the coordinates of the point that divides the interval AB with $A(1,4)$ and $B(5,2)$ externally in the ratio 1:3. 2
- c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{3x}$ 2
- d) Solve $\frac{4}{5-x} \geq 1$ 3
- e) Use the substitution $u = 1-x$ to evaluate $\int_{-1}^0 \frac{dx}{\sqrt{1-x}}$ 3

Question 2 (12 marks) Use a separate piece of paper

- a) Find $\frac{d}{dx}(x \cos^{-1} x)$ 2
- b) How many ten letter arrangements can be made using the letters of the word PHENOMENON? 2
- c) Write down the general solution of the equation $2 \sin \theta = \sqrt{3}$ 2
- d) State the domain and range of $y = 4 \cos^{-1}\left(\frac{x}{3}\right)$ and sketch the curve. 3
- e) Find the coefficient of x^7 in the expansion of $\left(x^2 - \frac{1}{x}\right)^{20}$ 3

Question 3 (12 marks) Use a separate piece of paper Marks

- a) Find $\int \cos^2 2x dx$ 2
- b) Prove $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{1}{\sqrt{10}} = \frac{\pi}{4}$ 3
- c) Use $x = 0.5$ to find an approximation for the root of $\cos x = x$ using one application of Newton's Method, correct to 2 decimal places. 3
- d) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $A \sin(x + \alpha)$ 2
- (ii) Hence solve $\sin x + \sqrt{3} \cos x = 1$ for $0 \leq x \leq 2\pi$ 2

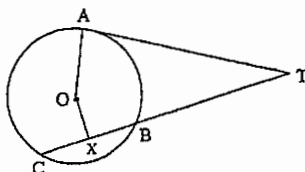
Question 4 (12 marks) Use a separate piece of paper

- a) Use the principle of mathematical induction to prove that; 3
- $$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$
- for all positive integers n .
- b) The two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$. 2
- (i) The equation of the tangent to $x^2 = 4ay$ at an arbitrary point $T(2at, at^2)$ on the parabola is $y = tx - at^2$. (You do not need to prove this)
- Show that the tangents at the points P and Q meet at R , where R is the point $\{a(p+q), apq\}$.
- (ii) If R lies on the line $y = -x - 5a$ find the relationship between p and q . 1
- (iii) Hence, or otherwise, find the locus of the midpoint of PQ . 2
- c) A molten plastic at a temperature of 250°C is poured into moulds to form car parts. After 20 minutes the plastic has cooled to 150°C . If the temperature after t minutes is $T^\circ\text{C}$, and if the temperature of the surroundings is 30°C , then the rate of cooling is approximately given by;
- $$\frac{dT}{dt} = -k(T - 30), \text{ where } k \text{ is a positive constant}$$
- (i) Verify that $T = 30 + 220e^{-kt}$ satisfies the above equation. 1
- (ii) Show that $k = \frac{1}{20} \log\left(\frac{11}{6}\right)$. 2
- (iii) The plastic can be taken out of the moulds when the temperature has dropped to 80°C . How long after the plastic has been poured will this temperature be reached? Give the answer to the nearest minute. 1

Question 5 (12 marks) Use a *separate* piece of paper

- a) Solve the equation $3x^3 - 17x^2 - 8x + 12 = 0$
given that the product of two of the roots is 4.

- b) A, B and C are three points on a circle centre O. The tangent at A meets BC produced at T. X is the midpoint of BC.



- (i) Prove that AOXT is a cyclic quadrilateral.
(ii) Hence state why $\angle AOT = \angle AXT$

- c) A particle moves in a straight line with an acceleration given by;

$$\frac{d^2x}{dt^2} = 9(x-2)$$

where x is the displacement in metres from the origin O after t seconds.
Initially the particle is 4 metres to the right of O with a velocity $v = -6$

- (i) Show that $v^2 = 9(x-2)^2$
(ii) Find an expression for v and hence find x as a function of t

Marks

3

3

1

2

3

Question 6 (12 marks) Use a *separate* piece of paper

- a) By considering both sides of the identity $(1+x)^m(1+x)^n = (1+x)^{m+n}$ and comparing coefficients, show that;

$$\binom{m+n}{3} = \binom{m}{3} + \binom{m}{2}\binom{n}{1} + \binom{m}{1}\binom{n}{2} + \binom{n}{3}$$

- b) A particle moves in a straight line and its position at time t seconds is given by

$$x = 5 + 4 \sin 2t$$

- (i) Show that the particle undergoes Simple Harmonic Motion
(ii) Find the centre and amplitude of the motion.
(iii) Determine the particle's maximum speed.

- c) The probability that a vaccine succeeds is $\frac{29}{30}$. An experiment is conducted m times with white mice.

- (i) What is the probability that the experiment will fail at least once?
(ii) If the probability that the experiment will fail at least once in m trials is greater than 90%, find the minimum number of times the experiment was conducted.

Marks

3

2

2

1

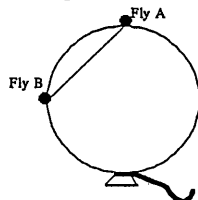
1

3

Question 7 (12 marks) Use a separate piece of paper

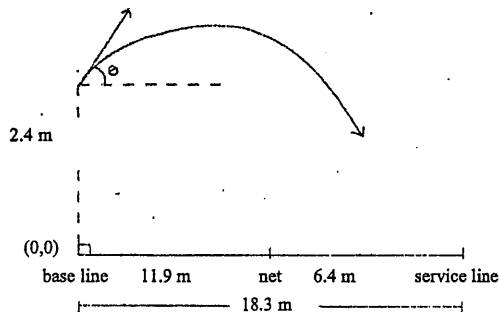
Marks

- a) Two flies are sitting on a spherical balloon of radius r cm, while it is being inflated at a constant rate of $5 \text{ cm}^3/\text{s}$. Assume that the balloon has no air in it to begin with and that the two flies are located at the North Pole and the Equator of the balloon.



- (i) Show that the distance between the two flies is $\sqrt{2}r$ cm. 1
- (ii) Hence show that the velocity of the two flies parting company is $\frac{5\sqrt{2}}{4\pi^2} \text{ cm/s}$ 2
- (iii) How fast are the two flies parting company after 3 seconds? Give your answer correct to two decimal places. 2
- b) In the 2006 Wimbledon Men's Final, Roger Federer's serve was measured to have an initial velocity of 180 km/h or 50 m/s.

Federer served the ball at the base line from a height of 2.4 metres at an angle of inclination of θ . In order not fault, the ball must land past the net and before the service line, that is a range between 11.9 metres and 18.3 metres.



Taking the origin as in the diagram and acceleration due to gravity as 9.8 m/s^2 ,

- (i) Derive the equations of motion and show that the position of the ball after t seconds is given by; 3
- $$x = 50t \cos \theta \quad \text{and} \quad y = -4.9t^2 + 50t \sin \theta + 2.4$$
- (ii) Hence show that $y = \frac{-4.9x^2 \sec^2 \theta}{50^2} + x \tan \theta + 2.4$ 2
- (iii) Calculate whether Federer will serve a fault if he serves the ball horizontally. 2

Question 4 (2)

a) Step 1: Prove true for $n=1$
 $LHS = \frac{1}{(3 \times 1)(2 \times 1)} = \frac{1}{6}$
 $RHS = \frac{1}{3 \times 1} = \frac{1}{3}$
 $\therefore LHS \neq RHS$
Hence the result is true for $n=1$.

Step 2: Assume true for $n=k$, where k is a positive integer.
 $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k-1)k} + \frac{1}{k(k+1)} = \frac{k}{k+1}$

Step 3: Prove true for $n=k+1$
 $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k-1)k} + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

Proof:
 $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k-1)k} + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$
 $= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$
 $= \frac{k(k+2) + 1}{(k+1)(k+2)}$
 $= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$
 $= \frac{(k+1)^2}{(k+1)(k+2)}$
 $= \frac{k+1}{k+2}$

Hence the result is true for $n=k+1$.
Hence the result is true for all n .

Question 3 (2)

a) $\int \cos 2x \, dx$
 $= \frac{1}{2} \int (\cos 4x) \, dx$
 $= \frac{1}{2} \left(\frac{\sin 4x}{4} \right) + c$
 $= \frac{1}{8} \sin 4x + c$

b) $x = \sin^{-1} \frac{1}{\sqrt{5}}$
 $\frac{1}{\sqrt{5}} = \sin x$
 $\frac{1}{\sqrt{5}} = \sin(x + \frac{\pi}{2})$
 $= \sin x \cos \frac{\pi}{2} + \cos x \sin \frac{\pi}{2}$
 $= 0 + \cos x$
 $\therefore \cos x = \frac{1}{\sqrt{5}}$
 $\therefore x = \cos^{-1} \frac{1}{\sqrt{5}}$

c) $f(x) = \cos x - x$
 $f'(x) = -\sin x - 1$
 $f(0.5) = 0.8776$
 $f'(0.5) = -1.4794$
 $x_1 = 0.5 - \frac{f(0.5)}{f'(0.5)}$
 $= 0.5 - \frac{0.8776}{-1.4794}$
 $= 0.76$ (to 2 d.p.)

d) $\sin x + \sqrt{3} \cos x$
 $= 2 \sin(x + \frac{\pi}{3})$
 $\therefore \sin(x + \frac{\pi}{3}) = \frac{1}{2}$
 $\sin \theta = \frac{1}{2}$
 $\theta = \frac{\pi}{6}$
 $x + \frac{\pi}{3} = \frac{\pi}{6}$
 $x = -\frac{\pi}{6}$

Question 2 (2)

a) $\frac{d}{dx} (x \cos x)$
 $= (x) \left(\frac{d}{dx} \cos x \right) + (\cos x) \left(\frac{d}{dx} x \right)$
 $= \frac{d}{dx} (x \cos x) + \cos x$
 $= \frac{d}{dx} (x \cos x) + \cos x$

b) Arrangement = $\frac{10!}{1! 3! 2! 2!}$
 $= \frac{10!}{1 \times 6 \times 2 \times 2}$
 $= \frac{10!}{24}$

c) $2 \sin \theta = \sqrt{3}$
 $\sin \theta = \frac{\sqrt{3}}{2}$
 $\theta = \pi/3$ or $(-\pi/3)$
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 $\theta = \pi/3$ or $(-\pi/3)$

d) $y = 4 \cos^2(\frac{\pi}{6})$
 $\text{domain: } -1 \leq \frac{\pi}{6} \leq 1$
 $\text{range: } 0 \leq \frac{\pi}{6} \leq \pi$
 $\frac{\pi}{6} \leq \frac{\pi}{6} \leq \pi$

e) $(x^2 - \frac{1}{x})^{20}$
 $T_{k+1} = {}^{20}C_k (x^2)^k (-\frac{1}{x})^{20-k}$
 $= {}^{20}C_k (-1)^k x^{40-3k}$
 $40-3k = 7$
 $3k = 33$
 $k = 11$

Question 1 (2)

a) $\int \frac{dx}{\sqrt{1-x^2}}$
 $= \int \frac{1}{\sqrt{1-x^2}} \, dx$
 $= \sin^{-1} x + c$

b) $A(1,4)$ $B(5,2)$
 $P = \left(\frac{3 \times 5 + 1 \times 2}{2}, \frac{3 \times 2 + 1 \times 4}{2} \right)$
 $= \left(\frac{16}{2}, \frac{10}{2} \right)$
 $= (8, 5)$

c) $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$
 $= \lim_{x \rightarrow 0} \frac{1}{1}$
 $= 1$

d) $\frac{1}{5-x} > 1$
 $\frac{1}{5-x} > 1$
 $\frac{1}{5-x} > 1$
 $\frac{1}{5-x} > 1$

e) $\int \frac{dx}{\sqrt{1-x^2}}$
 $= \sin^{-1} x + c$
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 $= \sin^{-1} x + c$

